

Hassan Mehmood

Web Development & Python Developer — Lahore, Pakistan

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SUMMARY

Motivated Computer Science graduate with hands-on experience in Python, JavaScript, HTML, CSS and MySQL. Passionate about web development and building real-world applications. Quick learner with strong problem-solving and communication skills, eager to contribute to a collaborative development team and grow with modern technologies.

SKILLS

Languages: Python, JavaScript, HTML, CSS

Databases: MySQL

Tools: Git, Google Gemini API, SpeechRecognition, pytsx3

Concepts: OOP, API Integration, Database Design

PROJECT EXPERIENCE

Jarvis - AI Voice Assistant

Python, Google Gemini API, SpeechRecognition, pytsx3

- Built a Python-based voice assistant powered by Google Gemini AI with wake-word activation and natural language understanding.
- Implemented speech-to-text and text-to-speech pipelines enabling real-time voice interaction and AI-generated responses.
- Integrated web browser control, music playback, and time/date query features using voice commands.
- Configured secure API key management using environment variables and a modular project structure for scalability.

Uber Backend - Ride-Sharing Application

Node.js, Express, MongoDB, Socket.IO, Redis, Stripe, JWT

- Developed a production-ready backend system for a ride-sharing platform with modular MVC architecture.
- Implemented JWT-based authentication, role-based access control, and security features including rate limiting, XSS protection, and NoSQL injection prevention.
- Built real-time ride tracking and driver-rider communication using Socket.IO WebSockets.
- Integrated Stripe for payments, Google Maps API for geolocation/routing, Twilio for SMS, and SendGrid for email notifications.
- Optimized database performance with MongoDB geospatial indexing and Redis caching for high-traffic scenarios.
- Wrote unit and integration tests using Jest and Supertest with comprehensive Swagger/OpenAPI documentation.

University Student Management System

MySQL / MariaDB

- Designed a 13-table normalized (3NF) MySQL schema covering departments, programs, students, courses, offerings and fees.
- Built 7 reporting views, 2 functions and 5 stored procedures, including prerequisite-checked enrollment and automatic fee-status updates.
- Added 3 triggers for capacity enforcement, grade-change auditing and fee date validation.
- Wrote and verified CRUD operations, multi-table joins, aggregate queries, subqueries and reporting scripts against realistic sample data.

EDUCATION

Associate's Degree in Computer Science

Superior University, Lahore

Oct 2023 - Jul 2025

Intermediate (Computer Science)

B.I.S.E Lahore

2023